

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 2 -- CTMC | M/M/1 Queue | Chapman-Kolmogorov & Diagonalization

Duration: 90 minutes. Total: 100 points. Show all working.

Notation: $Q(i,j)$ = rate from state i to state j in a CTMC. $q(i) = -Q(i,i)$ = total leaving rate from state i . $\rho = \lambda/\mu$ = traffic intensity. $P^n(i,j)$ = n -step transition probability.

Q1 (30 points). A server switches between Idle (state 0), Busy (state 1), and Overloaded (state 2). The rate matrix is:

	0	1	2
0	[-2.0	1.5	0.5]
1	[1.0	-3.0	2.0]
2	[0.5	1.5	-2.0]

- Verify that each row sums to 0. Then find $q(0)$, $q(1)$, $q(2)$ (the total leaving rate from each state) and state the distribution of the holding time $H(i)$ in each state.
 - Write the global balance equations $\pi \cdot Q = 0$ for this chain and solve for the stationary distribution $\pi(0)$, $\pi(1)$, $\pi(2)$.
 - Check whether detailed balance holds between states 0 and 2: compute $\pi(0) \cdot Q(0,2)$ and $\pi(2) \cdot Q(2,0)$ using your answer from (b).
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Q2 (35 points). A single-server queue (M/M/1) has Poisson arrivals at rate $\lambda = 5$ customers per hour and exponential service at rate $\mu = 8$ customers per hour.

- Compute $\rho = \lambda/\mu$ and verify the queue is stable. Find $\pi(0) = P(\text{server idle})$ and $\pi(n)$ for general n . What is $P(\text{more than 3 customers in system})$?
 - Compute $E[N]$ (mean number in system) and use Little's law $E[N] = \lambda \cdot E[T]$ to find $E[T]$ (mean time in system). Then find $E[T_q] = E[T] - 1/\mu$ (mean waiting time in queue).
 - The server takes a short break when the queue empties. This creates an M/M/1 variant where the server only starts work when there are at least 2 customers waiting. Would you expect $E[N]$ to increase or decrease compared to the standard M/M/1? Give a qualitative explanation (no calculation needed).
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Q3 (35 points). A Markov chain on $\{0, 1\}$ has transition matrix:

	0	1
0	[0.6	0.4]
1	[0.3	0.7]

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- a) Use the Chapman-Kolmogorov equation $P^{(2)} = P * P$ to compute $P^2(0, 0)$ directly (the probability of being in state 0 after 2 steps, starting from state 0).
- b) Find the two eigenvalues of P . Recall that for a 2×2 stochastic matrix with $P(0,1) = p$ and $P(1,0) = q$, the eigenvalues are $\lambda(1) = 1$ and $\lambda(2) = 1 - p - q$.
- c) Using the diagonalization formula for the n -step probability:

$$P^n(0,0) = q/(p+q) + \lambda(2)^n * p/(p+q)$$

where $p = P(0,1) = 0.4$ and $q = P(1,0) = 0.3$, compute $P^3(0,0)$ and find the limit as $n \rightarrow \infty$.